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Student Number

## 2016/17

EXAMINATION 1

# Extension 2 Mathematics

### General Instructions

- Reading time – 5 minutes
- Working time - 1 hours
- Write using blue or black pen  
Black pen is preferred
- Approved calculators may be used
- The 2 unit / Extension 1 formulae sheet is provided with this paper.
- In Questions 5-6 show relevant mathematical reasoning and/or calculations
- Start a new booklet for each question

**Total Marks – 40**

### Section I - Page 4

**4 marks**

- Attempt Questions 1 – 4
- Allow about 6 minutes for this section

### Section II - Pages 5-8

**36 marks**

- Attempt Questions 5 – 7
- Allow about 54 minutes for this section

| Question     | Marks      |
|--------------|------------|
| 1 - 4        | /4         |
| 5            | /14        |
| 6            | /11        |
| 7            | /11        |
| <b>Total</b> | <b>/40</b> |

**THIS QUESTION PAPER MUST NOT BE REMOVED FROM THE EXAMINATION ROOM**

*This assessment task constitutes 15% of the Higher School Certificate Course Assessment*

## Section I

4 marks

Attempt Questions 1 – 4

Allow about 6 minutes for this section

Use the multiple-choice answer sheet for questions 1 – 4 (Detach from paper)

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Q1

What is the value of  $\frac{3}{iz}$  if  $z = 1 - i$ ?

(A)  $-3 - i$

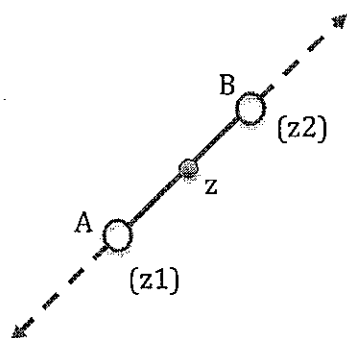
(B)  $3 - 3i$

(C)  $\frac{-3-3i}{2}$

(D)  $\frac{3-3i}{2}$

Q2

The locus of  $z$  is the line segment AB, excluding the points A and B.



Which of the following is true?

(A)  $\arg(z - z_1) - \arg(z - z_2) = 0$

(B)  $\arg(z_2 - z_1) - \arg(z - z_1) = \pi$

(C)  $\arg\left(\frac{z-z_1}{z-z_2}\right) = \pi$

(D)  $\arg(z_1 - z) = \arg(z_2 - z)$

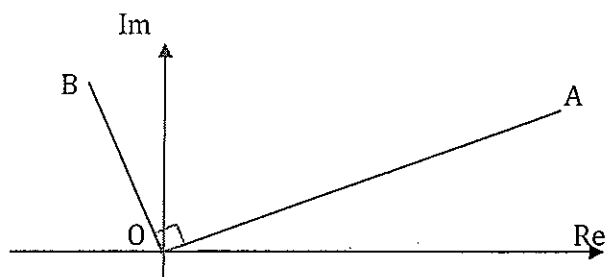
Q3

If  $n^2 \equiv an(n+1) + b(n-1) + c$ , then the value of the constants  $a$ ,  $b$  and  $c$  is:

- (A)  $a=1, b=1, c=1$
- (B)  $a=1, b=-1, c=1$
- (C)  $a=1, b=1, c=-1$
- (D)  $a=1, b=-1, c=-1$

Q4

In the Argand diagram below, the points A and B represent the complex numbers  $z$  and  $\omega$  respectively, where  $\angle AOB=90^\circ$ .



If  $|\overrightarrow{OA}| = 2|\overrightarrow{OB}|$  which of the following is correct?

- (A)  $\omega = 2iz$
- (B)  $\omega = -2iz$
- (C)  $\omega = -\frac{iz}{2}$
- (D)  $\omega = -\frac{z}{2i}$

## Section II

36 marks

Attempt Questions 5-7

Allow about 54 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 4 – 5, your responses should include relevant mathematical reasoning and/or calculations.

**Question 5** (14 marks) Use a SEPARATE writing booklet

|     |                                                                                                                                                                                                                                                                                                                                    |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| (a) | <p>The polynomial equation <math>x^3 + 3x^2 - 2x - 1 = 0</math> has roots <math>\alpha, \beta, \gamma</math>.</p> <p>Find the polynomial equation whose roots are <math>2\alpha, 2\beta, 2\gamma</math> ?</p>                                                                                                                      | 2 |
| (b) | <p>A polynomial equation <math>P(x)</math> has remainder 6 when divided by <math>x + 2</math>, and remainder 9 when divided by <math>x - 1</math>.</p> <p>What is the remainder when <math>P(x)</math> is divided by <math>(x + 2)(x - 1)</math> ?</p>                                                                             | 3 |
| (c) | <p>The polynomial equation <math>P(x) = ax^4 + bx^3 + cx^2 + e</math> has remainder <math>-2</math> when divided by <math>x + 1</math> and a double root at <math>x = 1</math>.</p> <p>Show that <math>4a + 2c = -3</math></p>                                                                                                     | 3 |
| (d) | <p>Let <math>P(z) = z^4 + 2kz^3 + 2k^2z^2 + 2kz + 1</math>, where <math>k</math> is real.</p> <p>Let <math>\beta = x + iy</math>, where <math>x</math> and <math>y</math> are real.</p> <p>Suppose that <math>\beta</math> and <math>i\beta</math> are zeros of <math>P(z)</math>, where <math>\bar{\beta} \neq i\beta</math>.</p> |   |
|     | i) Explain why $\bar{\beta}$ and $-i\bar{\beta}$ are zeros of $P(z)$                                                                                                                                                                                                                                                               | 1 |
|     | ii) Show that $P(z) = z^2(z + k)^2 + (kz + 1)^2$                                                                                                                                                                                                                                                                                   | 1 |
|     | iii) Hence show that if $P(z)$ has a real zero then<br>$P(z) = z^2(z - 1)^2 + (1 - z)^2$ OR $P(z) = (z^2 + 1)(z + 1)^2$ .                                                                                                                                                                                                          | 2 |
|     | iv) Show that all zeros of $P(z)$ have modulus 1.                                                                                                                                                                                                                                                                                  | 2 |

**End of Question 5**

**Question 6 (11 marks)** Use a SEPARATE writing booklet

|                          |                                                                                                                  |   |
|--------------------------|------------------------------------------------------------------------------------------------------------------|---|
| (a)                      | i) Find the square roots of $-8 + 6i$ .                                                                          | 2 |
|                          | ii) Solve $z^2 - (3 - 5i)z - 2 - 9i = 0$ .                                                                       | 2 |
|                          | iii) Sketch the locus of $z$ , if<br>$4 z - 2 + i  =  z^2 - (3 - 5i)z - 2 - 9i $                                 | 3 |
| (b)                      | (i) Solve the equation $z^5 + 32 = 0$ and list down all the roots in mod-arg form.                               | 2 |
|                          | (ii) Hence express $z^4 - 2z^3 + 4z^2 - 8z + 16$ as the product of two quadratic factors with real coefficients. | 2 |
| <b>End of Question 6</b> |                                                                                                                  |   |
|                          |                                                                                                                  |   |

**Question 7 (11 marks)** Use a SEPARATE writing booklet

(a)

Points A, B, C and D on an Argand diagram representing the complex numbers  $-2i$ ,  $4$ ,  $2 + 4i$  and  $2i$  respectively. ABCD form a convex quadrilateral as shown below. Squares are drawn outside the quadrilateral on each of the four sides. P, Q, R and S are the centres of the squares on AB, BC, CD and AD respectively.

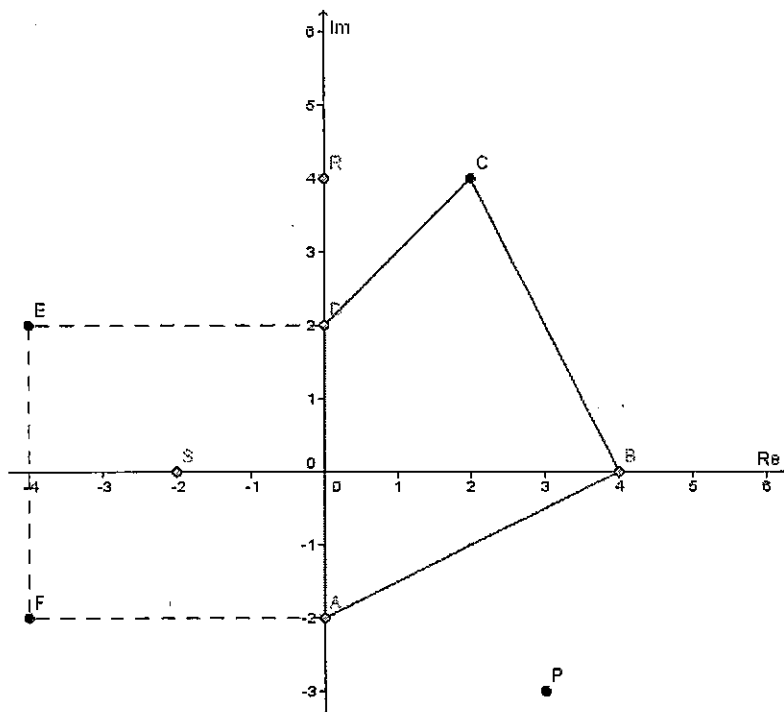
- i) Using vector representation of complex numbers, show that the complex number represented by Q is  $5 + 3i$ .

3

(It is given that P, R and S represent  $3 - 3i$ ,  $4i$  and  $-2$  respectively. You don't need to prove this)

- ii) Hence prove that PR is equal and perpendicular to QS, using vector representation of complex numbers.

3



(b) If  $z = \cos\theta + i\sin\theta$ ,

- (i) Show that  $z^n + \frac{1}{z^n} = 2\cos(n\theta)$  and  $z^n - \frac{1}{z^n} = 2i\sin(n\theta)$ .

2

- (ii) Hence show that  $\cos^4\theta + \sin^4\theta = \frac{1}{4}(\cos 4\theta + 3)$

3

**End of Question 7**

**End of Examination ☺**



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Student Number

# Mathematics

## Section I – Multiple Choice Answer Sheet

Use this multiple-choice answer sheet for questions 1 – 4. Detach this sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample:**       $2 + 4 =$       (A) 2      (B) 6      (C) 8      (D) 9  
                                  A ☐      B ☒      C ☐      D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A  B  C  D 

**Start Here** →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☒ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐

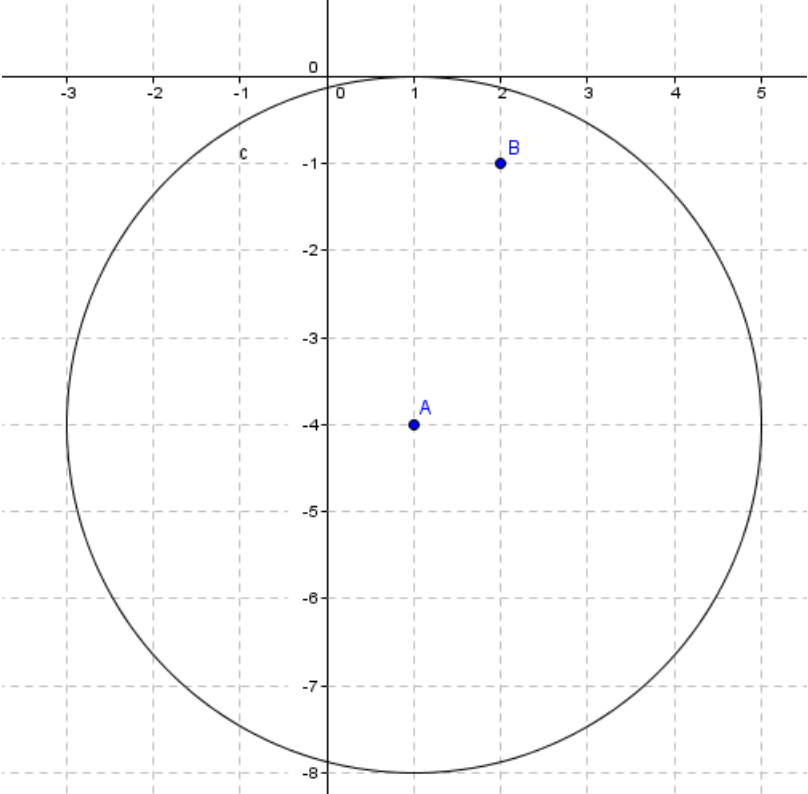
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Extension 2 Task 1 2017 ANSWERS

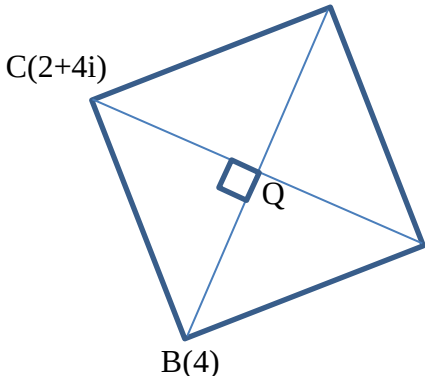
| Q | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks                                        |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| 1 | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                            |
| 2 | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                            |
| 3 | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                            |
| 4 | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                                            |
| 5 | <p>a) Let <math>P(x) = x^3 + 3x^2 - 2x - 1 = 0</math> have roots <math>\alpha, \beta, \gamma</math>.<br/> <math>\alpha</math> is a root of <math>P(x) \rightarrow P(\alpha) = 0</math><br/> When<br/> <math>2\alpha = x</math>,<br/> <math>\alpha = \frac{x}{2} \rightarrow P\left(\frac{x}{2}\right) = 0</math><br/> is the polynomial equation whose roots are <math>2\alpha, 2\beta, 2\gamma</math><br/> <math>P\left(\frac{x}{2}\right) = \left(\frac{x}{2}\right)^3 + 3\left(\frac{x}{2}\right)^2 - 2\left(\frac{x}{2}\right) - 1 = 0</math><br/> i.e. <math>x^3 + 6x^2 - 8x - 8 = 0</math></p> <p>b) When <math>P(x)</math> is divided by <math>(x+2)</math> the remainder is <math>P(-2)</math> by remainder theorem.<br/> <math>\therefore P(-2) = 6</math><br/> Similarly <math>P(1) = 9</math><br/> Since <math>(x+2)(x-1)</math> is a quadratic polynomial, the remainder when <math>P(x)</math> is divided by <math>(x+2)(x-1)</math> is a linear function <math>(ax+b)</math> and let the quotient function be <math>Q(x)</math>.<br/> <math>\therefore P(x) \equiv (x+2)(x-1).Q(x) + (ax+b)</math><br/> This division transformation is an identity<br/> i.e. the statement is true for all values of <math>x</math>.</p> <p><math>\therefore</math> when<br/> <math>x = -2 \quad P(-2) = -2a + b = 6 \quad (1)</math><br/> <math>x = 1 \quad P(1) = a + b = 9 \quad (2)</math><br/> Solving equations (1) and (2) simultaneously gives <math>a = 1, b = 8</math><br/> <math>\therefore</math> the required remainder is <math>(x+8)</math></p> | <p>1</p> <p>1</p> <p>1</p> <p>1</p> <p>1</p> |

| Q | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5 | <p>c) By remainder theorem when <math>P(x)</math> is divided by <math>(x + 1)</math> the remainder is <math>P(-1)</math>.</p> $P(-1) = -2 \rightarrow a(-1)^4 + b(-1)^3 + c(-1)^2 + e = -2$ $\rightarrow a - b + c + e = -2 \quad (1)$ <p><math>P(x)</math> has a double root at <math>x = 1 \rightarrow</math></p> $P(1) = 0 \rightarrow a + b + c + e = 0 \quad (2)$ <p>And as <math>P'(x) = 4ax^3 + 3bx^2 + 2cx = 0</math></p> $P'(1) = 0 \rightarrow 4a + 3b + 2c = 0 \quad (3)$ $(2) - (1) \rightarrow 2b = 2 \rightarrow b = 1$ <p>Sub <math>b = 1</math> into (3)</p> $4a + 3 + 2c = 0$ $4a + 2c = -3$ <p>d)</p> <p>i) Complex conjugate of <math>\beta = \bar{\beta}</math></p> <p>Complex conjugate of <math>i\bar{b} = \overline{i\bar{b}} = i\bar{b} = -i\bar{b}</math></p> <p><math>P(z) = z^4 + 2kz^3 + 2k^2z^2 + 2kz + 1</math>, where <math>k</math> is real, and the coefficients of <math>P(z)</math> are real</p> <p><math>\rightarrow \beta</math> is a root, implies <math>\bar{\beta}</math> is a root of <math>P(z)</math> by conjugate root theorem</p> <p>Similarly <math>i\beta</math> is a root, implies <math>-i\bar{\beta}</math> is a root of <math>P(z)</math></p> <p>ii) <math display="block">P(z) = z^4 + 2kz^3 + 2k^2z^2 + 2kz + 1</math> <math display="block">= (z^4 + 2kz^3 + k^2z^2) + (k^2z^2 + 2kz + 1)</math> <math display="block">= z^2(z + k)^2 + (kz + 1)^2</math></p> <p>iii) <math display="block">P(z) = z^2(z + k)^2 + (kz + 1)^2</math> <p>As <math>P(z)</math> is the sum of two complete squares, for <math>P(z)</math> to be zero, both <math>z + k</math> and <math>kz + 1</math> must be zero.</p> <p>Solving <math>z + k = 0</math> and <math>kz + 1 = 0</math> gives <math>k = \pm 1</math>.</p> <p>When <math>k = 1, P(z) = z^2(z + 1)^2 + (z + 1)^2</math></p> <math display="block">= (z + 1)^2(z^2 + 1) \text{ ---- (1) (by factorising)}</math> <p>When <math>k = -1, P(z) = z^2(z - 1)^2 + (1 - z)^2 \text{ ---- (2)}</math></p> <p>iv) From (iii) above, when <math>P(z) = 0</math>, gives <math>z = -1</math> and <math>\pm i</math> from (1) and <math>z = 1</math> from (2)</p> <p>ie. The roots of <math>P(z) = 0</math> are <math>1, -1, i</math> and <math>-i</math></p> <p>The modulus of each of these roots are 1.</p> </p> | <p>1</p> <p>1 mark for using <math>P(1) = 0</math> and <math>P'(1) = 0</math></p> <p>1 mark for using the equations to get the required answer</p> <p>1 mark for stating the conjugate theorem and that all coefficients are real</p> <p>1 mark for correct working</p> <p>1 mark for getting the two values of <math>k</math></p> <p>1 mark for getting the two required expressions for <math>P(z)</math>.</p> <p>1 mark for each for using the two equations to get the values of <math>z</math></p> <p>1 mark for stating the modulus is 1</p> |

| Q | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks                                                                     |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
| 6 | <p>a)</p> <p>i) Students may do by a number of methods<br/>(by calculating using shorthand techniques)<br/>Go straight to<br/> <math display="block">-8 + 6i = (1 + 3i)^2</math> <math display="block">\therefore \sqrt{-8 + 6i} = \pm(1 + 3i)</math></p> <p>OR long way<br/> <math display="block">\sqrt{-8 + 6i} = x + iy</math> <math display="block">\therefore -8 + 6i = x^2 - y^2 + 2ixy</math> Equating coefficients of real and imaginary parts<br/> <math display="block">-8 = x^2 + y^2 \quad (1)</math> <math display="block">6i = 2ixy</math> <math display="block">3 = xy</math> <math display="block">y = \frac{3}{x} \quad (2)</math></p> <p>Sub. (2) into (1)</p> $-8 = x^2 - \left(\frac{3}{x}\right)^2$ $-8x^2 = x^4 - 9$ $x^4 + 8x^2 - 9$ $(x^2 + 9)(x^2 - 1) = 0$ $\therefore x = \pm 1$ sub. into (2)<br>$y = \pm 3$ $\therefore \sqrt{-8 + 6i} = \pm(1 + 3i)$ | 2 marks for getting the two square roots correctly by any relevant method |

| Q | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks                               |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 6 | <p>a/</p> <p>ii) <math>z^2 - (3 - 5i)z - 2 - 9i = 0</math><br/> By quadratic formula<br/> <math display="block">z = \frac{(3 - 5i) \pm \sqrt{(3 - 5i)^2 + 4(2 + 9i)}}{2}</math> <math display="block">z = \frac{(3 - 5i) \pm \sqrt{-8 + 6i}}{2}</math> <math display="block">z = \frac{(3 - 5i) \pm (1 + 3i)}{2} \quad \text{from part i)}</math> <math display="block">z = (2 - i), (1 - 4i)</math></p> <p>iii) <math>4 z - 2 + i  =  z^2 - (3 - 5i)z - 2 - 9i </math><br/> <math>\rightarrow 4 z - 2 + i  =  [z - (2 - i)][z - (1 - 4i)]  \quad \text{from part ii}</math><br/> <math>\rightarrow 4 z - 2 + i  =  z - 2 + i  z - 1 + 4i </math><br/> <math>\rightarrow  z - 2 + i ( z - 1 + 4i  - 4) = 0</math></p> <p>When <math> z - 2 + i  = 0 \rightarrow</math> the locus of <math>z</math> is just the point <math>(2 - i)</math></p> <p>When <math> z - 1 + 4i  = 4 = 0 \rightarrow</math> the locus of <math>z</math> is the circle with centre <math>(1 - 4i)</math> and radius 4.</p> <p>Sketch is shown below:</p>  | <p>2</p> <p>1</p> <p>1</p> <p>1</p> |

| Q | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                                                                                                                                                             |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6 | <p>b) i)</p> $z^5 = -32$ $z^5 = 2^5(-1 + 0i)$ $z^5 = 2^5 \operatorname{cis}(2k - 1)\pi \text{ where } k \text{ is an integer}$ $z = 2 \operatorname{cis}(2k - 1)\frac{\pi}{5} \text{ (By De Moivre's Theorem)}$ <p>The roots of <math>z^5 + 32 = 0</math> are:</p> <p>when <math>k = 0</math>, <math>z = 2 \operatorname{cis}\left(-\frac{\pi}{5}\right) = z_1</math></p> <p>when <math>k = 1</math>, <math>z = 2 \operatorname{cis}\left(\frac{\pi}{5}\right) = \bar{z}_1</math></p> <p>when <math>k = -1</math>, <math>z = 2 \operatorname{cis}\left(-\frac{3\pi}{5}\right) = z_2</math></p> <p>when <math>k = 2</math>, <math>z = 2 \operatorname{cis}\left(\frac{3\pi}{5}\right) = \bar{z}_2</math></p> <p>when <math>k = -2</math>, <math>z = 2 \operatorname{cis}(\pi) = -2</math></p> <p>ii)</p> $\frac{z^5 + 32}{z + 2} = (z + 2)[(z - z_1)(z - \bar{z}_1)][(z - z_2)(z - \bar{z}_2)]$ $\frac{z^5 + 32}{z + 2} = [z^2 - (z_1 + \bar{z}_1)z + z_1 \bar{z}_1][z^2 - (z_2 + \bar{z}_2)z + z_2 \bar{z}_2]$ $\frac{z^5 + 32}{z + 2} = \left[z^2 - 4 \cos \frac{\pi}{5} z + 4\right] \left[z^2 - 4 \cos \frac{3\pi}{5} z + 4\right]$ | <p>2<br/>for all<br/>correct<br/>roots</p> <p>1<br/>for<br/>factorising<br/>and<br/>grouping<br/>the<br/>conjugates</p> <p>1<br/>for the<br/>final<br/>answer</p> |

| Q | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks                               |
|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 7 | <p>a)</p>  <p>i)</p> <p>In right angled <math>\Delta BQC</math>, <math>\frac{CQ}{CB} = \cos \frac{\pi}{4}</math><br/> <math>ie. CQ = \frac{1}{\sqrt{2}} CB</math></p> <p><math>\overrightarrow{CQ}</math> is obtained by rotating <math>\frac{1}{\sqrt{2}} \overrightarrow{CB}</math> in the anti-clockwise sense by <math>\frac{\pi}{4}</math> radians about <math>C</math>.</p> <p><math>\therefore \overrightarrow{CQ} = \frac{1}{\sqrt{2}} \overrightarrow{CB} \text{ cis } \frac{\pi}{4}</math></p> <p><math>\overrightarrow{CB} = \overrightarrow{CO} + \overrightarrow{OB}</math><br/> <math>= (-2 - 4i) + 4</math><br/> <math>= 2 - 4i</math></p> <p><math>\therefore \overrightarrow{CQ} = \frac{1}{\sqrt{2}} (2 - 4i) \text{ cis } \frac{\pi}{4}</math><br/> <math>= (1 - 2i)(1 + i)</math><br/> <math>= 3 - i</math></p> <p><math>\overrightarrow{OQ} = \overrightarrow{OC} + \overrightarrow{CQ}</math><br/> <math>= (2 + 4i) + (3 - i)</math><br/> <math>= (5 + 3i)</math></p> <p><math>\therefore Q</math> represents <math>(5 + 3i)</math>.</p> <p>ii)</p> <p><math>\overrightarrow{PR} = \overrightarrow{PO} + \overrightarrow{OR}</math><br/> <math>= (-3 + 3i) + 4i</math><br/> <math>= -3 + 7i</math><br/> <math>\overrightarrow{QS} = -2 - (5 + 3i)</math><br/> <math>= (-7 - 3i)</math><br/> <math>i\overrightarrow{PR} = i(-3 + 7i)</math><br/> <math>= (-7 - 3i)</math><br/> <math>= \overrightarrow{QS}</math></p> | <p>1</p> <p>1</p> <p>1</p> <p>1</p> |

|   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                  |
|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <p>Since <math>\overrightarrow{QS} = i\overrightarrow{PR}</math>, <math> \overrightarrow{QS}  =  i\overrightarrow{PR} </math> and <math>\arg(\overrightarrow{QS}) = \arg(i\overrightarrow{PR})</math>.</p> <ul style="list-style-type: none"> <li><math> \overrightarrow{QS}  =  i\overrightarrow{PR} </math></li> </ul> $ \overrightarrow{QS}  =  i  \overrightarrow{PR} $ $=  \overrightarrow{PR} $ <p><b>ie. the lengths of QS and PR are equal.</b></p> <ul style="list-style-type: none"> <li><math>\arg(\overrightarrow{QS}) = \arg(i\overrightarrow{PR})</math></li> <li><math>\arg(\overrightarrow{QS}) = \arg(i) + \arg(\overrightarrow{PR})</math></li> </ul> $= \frac{\pi}{2} + \arg\overrightarrow{PR}$ <p><b>ie. QS and PR are perpendicular</b></p> <p><b>Therefore QS and PR are both equal and perpendicular.</b></p> | <p>1</p> <p>1</p>                                                                                                                                                                |
| 7 | <p>b (i)</p> $z = \cos\theta + i\sin\theta$ $z^n = \cos(n\theta) + i\sin(n\theta) \quad (1) \quad \text{By De Moivre's Theorem}$ $\frac{1}{z^n} = z^{-n}$ $= \cos(-n\theta) + i\sin(-n\theta) \quad \text{By De Moivre's Theorem}$ $= \cos(n\theta) - i\sin(n\theta) \quad (2)$ <p>(1) + (2) gives <math>z^n + \frac{1}{z^n} = 2\cos(n\theta)</math></p> <p>(1) - (2) gives <math>z^n - \frac{1}{z^n} = 2i\sin(n\theta)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                    | <p>1</p> <p>Mark for using De Moivre's theorem Correctly</p> <p>1 mark for getting the expressions for <math>z^n + \frac{1}{z^n}</math> and <math>z^n - \frac{1}{z^n}</math></p> |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                       |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p>ii)</p> $2 \cos \theta = z + \frac{1}{z}$ $\therefore 16 \cos^4 \theta = \left( z + \frac{1}{z} \right)^4 \quad (1)$ $2i \sin \theta = z - \frac{1}{z}$ $\therefore 16 \sin^4 \theta = \left( z - \frac{1}{z} \right)^4 \quad (2)$ $(1)+(2) \quad 16(\cos^4 \theta + \sin^4 \theta) = \left( z + \frac{1}{z} \right)^4 + \left( z - \frac{1}{z} \right)^4$ $= 2 \left( z^4 + \frac{1}{z^4} \right) + 12$ $= 2 \cos 4\theta + 12$ $\therefore \cos^4 \theta + \sin^4 \theta = \frac{1}{8} [\cos 4\theta + 3]$ | <p>1 mark for getting expressions for <math>\cos^4 \theta</math> and <math>\sin^4 \theta</math></p> <p>1 mark for simplifying</p> <p>1 mark for using De Moivre's theorem again to get the answer</p> |
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